

DOCUMENT 6: IMPLEMENTATION & FEASIBILITY PLAN

Project Name: PARIKSHAK-AI (परीक्षक-AI)
Challenge: Challenge 03 — AI-Driven Examination & On-Screen Marking Transformation
Track: Technical Track | **Team Name:** eMitra

1. Feasibility Assessment: Non-Invasive Bolt-On Architecture

The primary reason previous digital evaluation tenders in Madhya Pradesh failed (such as Barkatullah University's cancelled ₹8 Crore RFP and DAVV Indore's frozen pilot) was their insistence on monolithic overhauls—attempting to replace existing university databases, examination registers, and scanning logistics all at once.

PARIKSHAK-AI is engineered with a **Non-Invasive Microservice Architecture**. It does not replace MPOnline's established ASP.NET Core portal or Oracle 19c database at the MP State Data Centre (SDC) in Bhopal. Instead, it operates as an **Intelligent Cognitive Sidecar**:

- Ingests scanned booklet imagery via temporary, pre-signed HTTPS URLs.
- Performs offline layout analysis, question segmentation, and rubric alignment.
- Provides examiners with a responsive, low-bandwidth web evaluation canvas.
- Asynchronously commits validated marks directly into MPOnline's Tabulation Register via standard REST APIs and message queues.

2. Four-Phase Phased Rollout Roadmap

PARIKSHAK-AI DEPLOYMENT TIMELINE
[PHASE 1: PROTOTYPE & BENCHMARKING] → Oct 2026 – Dec 2026 • Hackathon prototype hardening • Corpus ingestion: 10,000 sample MP university answer scripts (Hindi/English) • CERT-In Empaneled Security VAPT & STQC Code Quality Audit
[PHASE 2: CONTROLLED UNIVERSITY PILOT] → Jan 2027 – Mar 2027 (Winter Exam) • Live 25,000-script pilot at DAVV Indore (MBA) or BU Bhopal (B.Tech) • Non-destructive V-cradle scanning deployment at Central Valuation Center • Calibration testing with 50 empanelled university professors
[PHASE 3: TOP-5 STATE UNIVERSITY EXPANSION] → Apr 2027 – Sep 2027 (Summer) • Rollout across DAVV, BU Bhopal, RGPV, Jiwaji, and Vikram University • ~50 Lakh answer scripts evaluated across 55 districts • Integration with MP State Data Centre (SDC) private GPU cluster
[PHASE 4: STATEWIDE INSTITUTIONAL SCALE] → 2028 Onwards • Full coverage across all 25 State Public Universities (3.0+ Crore booklets) • Integration with DigiLocker and National Academic Depository (NAD / ABC) • Citizen-centric governance: transparent result portals and diagnostic cards
[PHASE 5: NATIONAL EXPANSION & MULTI-LANGUAGE SCALE] → 2029 Onwards • Replicate to 3–5 high-demand states: UP, Rajasthan, Bihar, Maharashtra, CG • Extend multimodal vision to additional Indic scripts: Bangla, Tamil, Telugu • Serve non-university examination bodies: CBSE, State Boards, ICAI, Pharmacy • White-label SaaS offering through MPOnline/TCS iON partnership network • Pan-India TAM: 49 Crore scripts/year (AISHE 2022-23, ₹10,000+ Cr market)

3. Physical-to-Digital Scanning Protocol (The Anti-Tampering Standard)

To eliminate the physical security vulnerabilities that paused DAVV Indore’s digital evaluation rollout in 2026 (fear of guillotine spine-cutting, page-swapping, and unstitched sheet theft), PARIKSHAK-AI establishes a **Two-Tier Non-Destructive Scanning Standard**:

Tier 1: Non-Destructive Overhead V-Cradle Scanning (High-Stakes & Professional Exams)

- **Equipment:** High-speed overhead book scanners (e.g., CZUR M3000 Pro / Zeutschel OS 12002) equipped with V-shaped book cradles and auto-flattening curve algorithms.
- **Integrity:** The physical cotton thread binding (*Dhaga Silayi*) of the 36-page booklet remains **100% intact**. Not a single staple or thread is cut.
- **Throughput:** Scans an entire 36-page booklet in under 60 seconds at 300 DPI color/grayscale.

Tier 2: Leaf-Level 2D DataMatrix Validation (High-Volume General Exams)

- Where automatic document feeder (ADF) scanners are deployed for extreme volume (B.A. / B.Com. examinations):
- Every physical page features a pre-printed, serialized 2D DataMatrix code encoding both booklet UUID and sequential page number (BK-94812-P14).
- The scanning daemon enforces a strict **Discontinuity Trap**: if any page sequence is missing, skipped, or duplicated, the scanning batch halts instantly, and the physical packet is sequestered.
- Scanning operations occur strictly within the **University Strong Room (Gopneeya Shakha)** under dual-key access and continuous 360-degree CCTV surveillance.

4. Hardware Sizing & Sovereign Cloud Deployment

To guarantee total compliance with the **Digital Personal Data Protection (DPDP) Act 2023** and maintain 100% data residency within Madhya Pradesh:

- **Primary Hosting Facility:** MP State Data Centre (SDC), Bhopal.
- **Disaster Recovery (DR) Site:** TCS Tier-4 Data Centre / MeitY-empanelled CSP (AWS / Azure Central India).
- **Compute Sizing for Statewide Peak Load (500,000 Booklets / Day):**
- *Inference Cluster:* 4x Enterprise AI Nodes, each hosting 4x NVIDIA L40S 48GB GPUs (16 GPUs total) running quantized open-weights Indic Multimodal Vision models (e.g., Qwen2.5-VL / Sarvam Indic VLM) with vLLM PagedAttention.
- *Throughput target:* ~120 question evaluations per second (500,000 booklets a day over a 7-hour window) with headroom, to be confirmed by benchmarking the chosen model on this hardware.
- *Storage Infrastructure:* 150 TB encrypted NVMe-backed MinIO S3 object storage configured with write-once-read-many (WORM) immutability policies.

5. Risk Assessment & Comprehensive Mitigation Matrix

#	Identified Risk Domain	Severity	Operational / Technical Mitigation in PARIKSHAK-AI
1	Teacher Union Resistance (University Faculty Associations)	HIGH	Replace punitive screen lockouts with a 3-Tier Progressive Friction Protocol (Soft Nudge → Touchpoint Gate → Silent Shadow Review). Recommend raising examiner honorariums, funded partly from logistics savings.

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2	Rural Bandwidth Fluctuation	MEDIUM	Deploy Local Edge Caching Appliances (LECA) at District Lead Colleges (<i>Agrani Mahavidyalayas</i>) to pre-download daily bundles overnight, serving evaluators over local high-speed LAN without active internet dependencies.
3	Spine-Cutting / Page Swapping	HIGH	Mandate non-destructive V-cradle overhead scanners for professional courses; require leaf-level 2D serialized DataMatrix codes for ADF scanners.
4	Adversarial Prompt Injection	MEDIUM	Dual-turn sandboxed inference: Turn 1 extracts visual text into strict JSON schemas with zero command execution; Turn 2 evaluates within CDATA boundaries. Transcribed text pre-screened for injection signatures.
5	Low Paper GSM & Ink Bleed-Through	MEDIUM	Adaptive preprocessing pipeline applying Sauvola binarization and morphological background subtraction to eliminate reverse-page ink bleed-through on 54–60 GSM paper.
6	Displaced Revaluation Cash Flow	MEDIUM	Provide university Executive Councils with an estimated net operating balance (about +₹0.7 Crore a year for a 30-Lakh-script university) from reduced paper transport, escorts and litigation costs.

6. Regulatory & Statutory Certification Strategy

Prior to Phase 2 university pilot commissioning, PARIKSHAK-AI will undergo formal statutory certification:

- STQC Directorate Clearance:** Functional and performance testing for electronic governance applications under Ministry of Electronics and Information Technology (MeitY) guidelines.
- CERT-In Empaneled Security VAPT:** Rigorous vulnerability assessment and penetration testing verifying zero Broken Object Level Authorization (BOLA/IDOR) flaws.
- University Executive Council Resolutions:** Amending Ordinance No. 5 across participating universities to formally recognize digital assistive evaluation and Section 63 BSA electronic audit trails.

7. Commercial Sustainability & Revenue Model

A. SaaS Pricing Architecture (Per-Script Transaction Model)

PARIKSHAK-AI operates as a **zero-capex AI intelligence layer** atop existing OSM scanning infrastructure, charging a per-booklet transaction fee:

Component	Unit Cost	Volume (MP Statewide)	Annual Revenue
AI Copilot Inference (Gemini Flash, priced to cover current model cost)	₹3.00 / booklet	3.04 Crore scripts	₹9.12 Crore
Seed Calibration & Velocity Sentinel SaaS	₹1.50 / booklet	3.04 Crore scripts	₹4.56 Crore
Section 63 BSA Dossier Generation	₹0.50 / booklet	3.04 Crore scripts	₹1.52 Crore
Command Center Dashboard (Annual License)	₹15 Lakh / university	25 universities	₹3.75 Crore
Total MP Annual Revenue			₹18.95 Crore

B. Strategic Alignment with MPOnline's Revenue Expansion

MPOnline currently earns ~₹25 Crore/year from higher education form fees alone. The evaluation market (₹60–₹75 Crore/year for 3.04 Crore scripts across MP) remains **uncaptured revenue** lost to fragmented private vendor

tenders. PARIKSHAK-AI enables MPOnline to expand from upstream form collection into high-value downstream evaluation processing—doubling higher education revenue without additional kiosk infrastructure.

C. Funding via PM-USHA & RUSA Grants

MP state universities can seek infrastructure and IT-upgrade grants under PM-USHA (₹12,926 Crore national outlay), which can fund examination automation. PARIKSHAK-AI deployment qualifies as a **PM-USHA-compliant examination reform investment**, making the solution **self-funded from central government grants** rather than university operating budgets.